

Applications of Geospatial Technology in National Public Infrastructure



Kinmen Bridge (Opened 2022)

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2026.06.24



台灣世曦工程顧問股份有限公司
CECI Engineering Consultants, Inc., Taiwan

Outline



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About Us

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LiDAR Application

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National Maps

4

UAV Application

CHAPTER 1

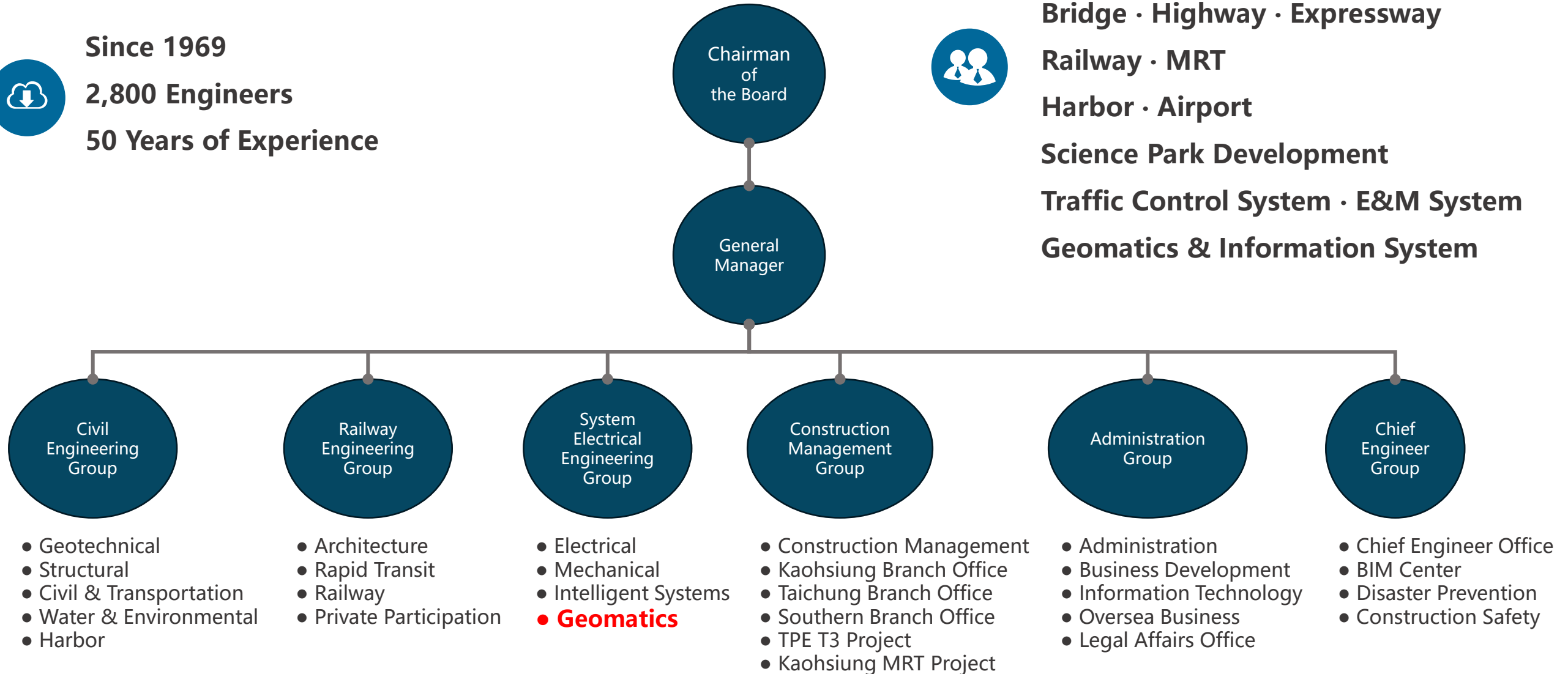
About Us



 Since 1969
2,800 Engineers
50 Years of Experience



Bridge · Highway · Expressway
Railway · MRT
Harbor · Airport
Science Park Development
Traffic Control System · E&M System
Geomatics & Information System



Global Engineering Program of CECI

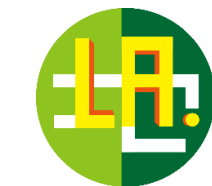
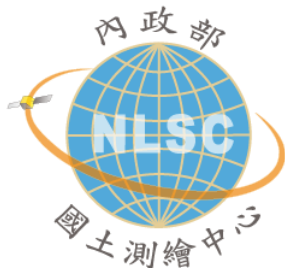




太魯閣國家公園



國土管理署



內政部地政司



桃機公司



交通部鐵道局



臺灣港務公司



臺北市政府



桃園市政府



花蓮縣政府



交通部高速公路局



交通部公路局



竹科管理局



中科管理局



南科管理局



新北市政府



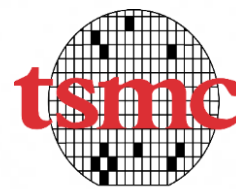
雲林縣政府



台北自來水公司



金門自來水廠



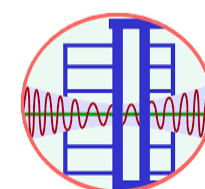
台灣積體電路



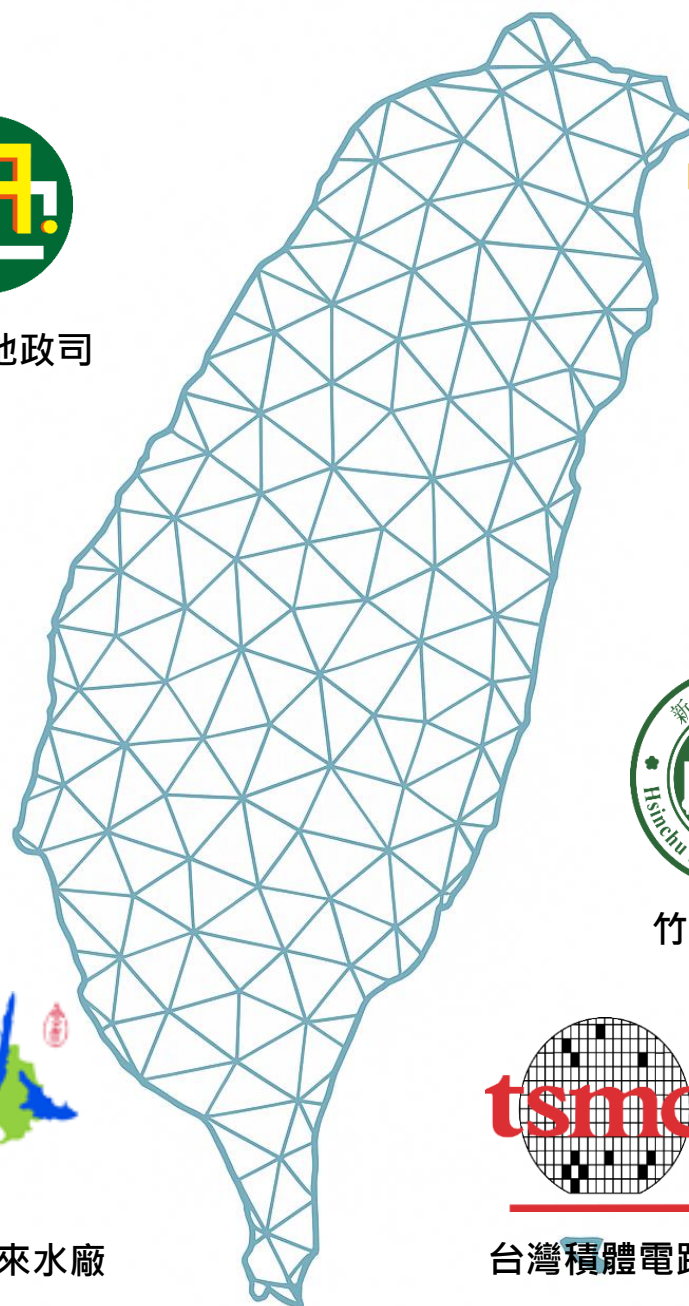
台灣電力公司



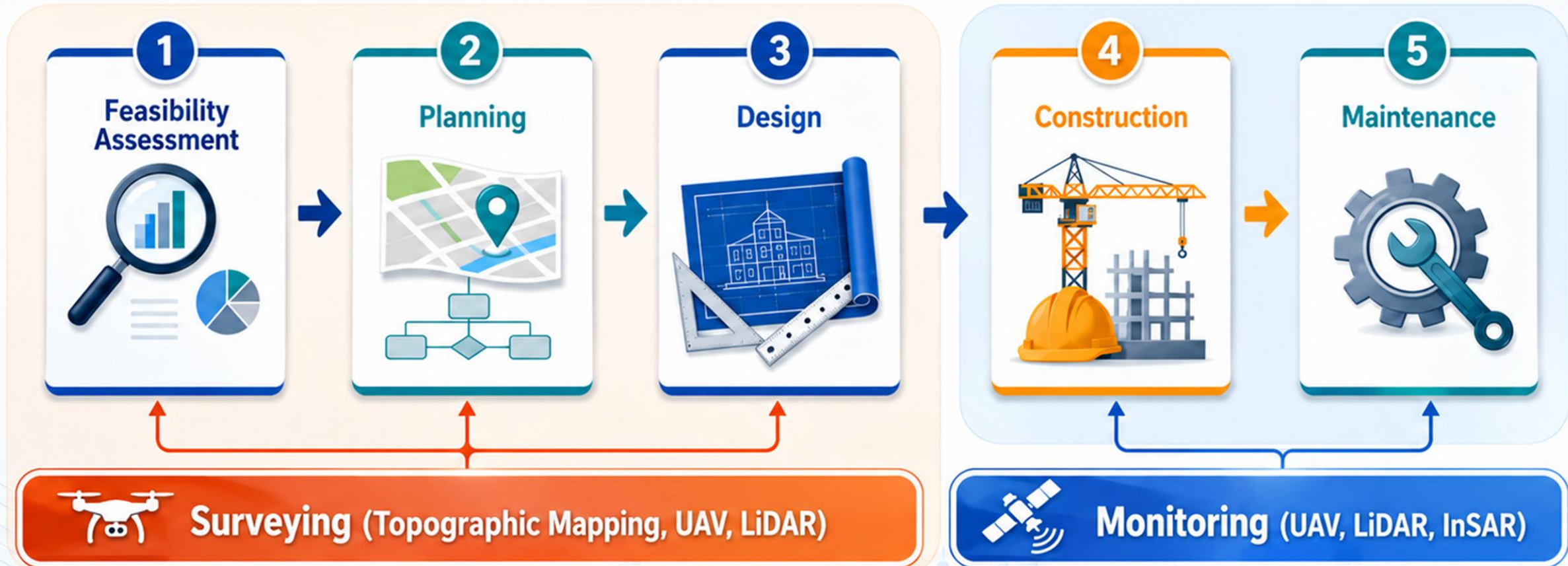
經濟部地礦中心



國家地震中心

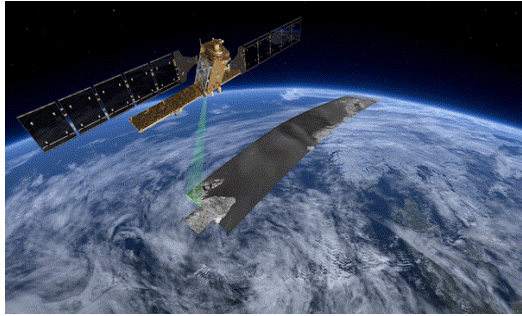


CECI Team

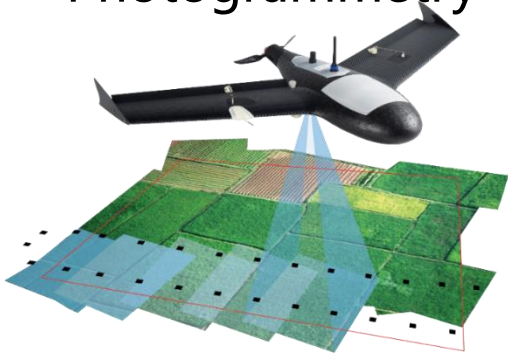


Geospatial Information Technology

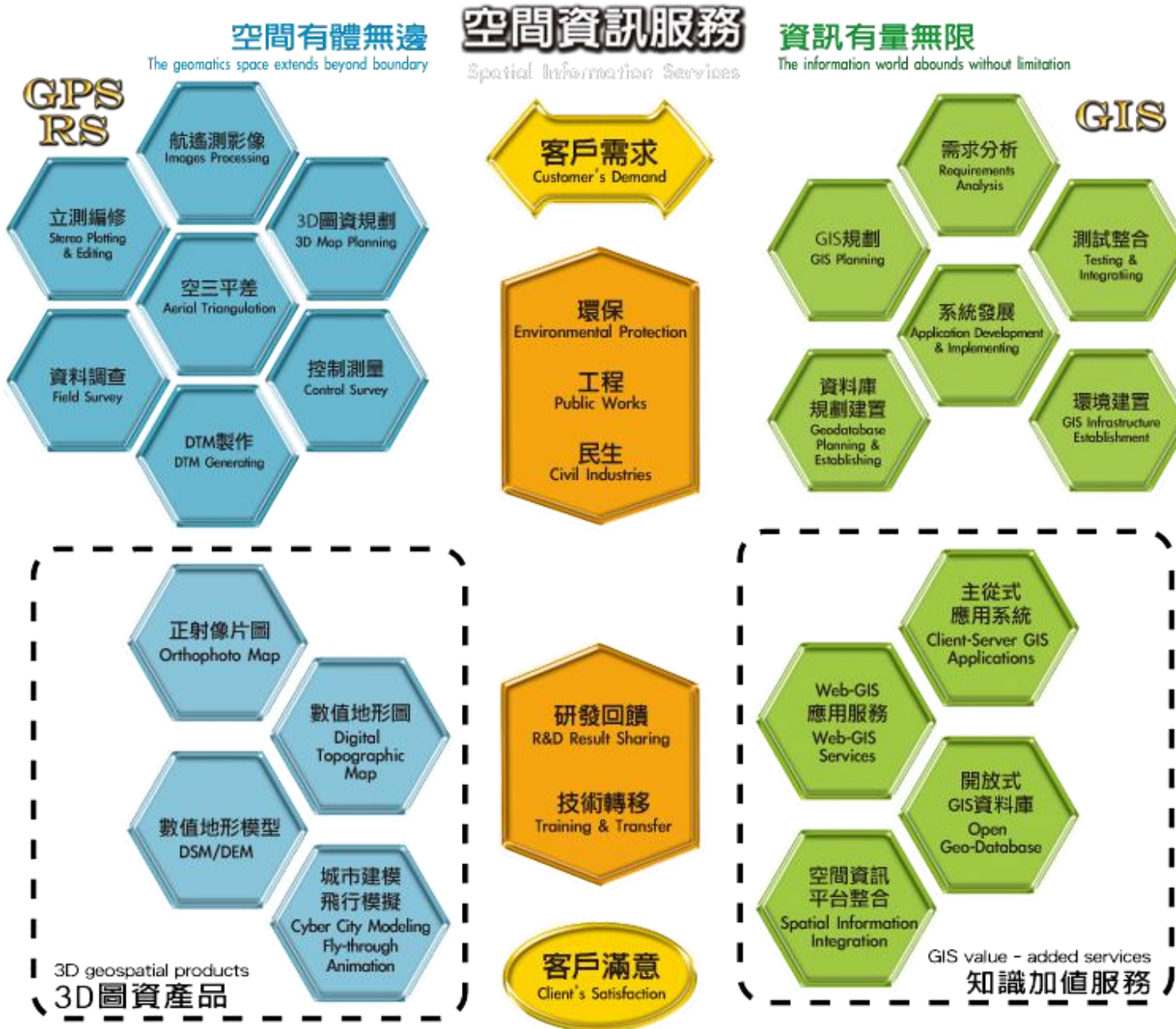
InSAR-Radar



Photogrammetry



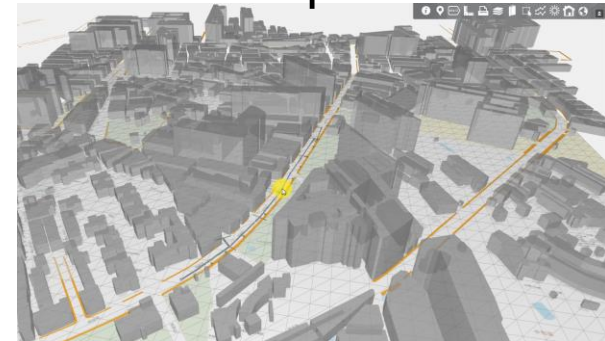
Mobile LiDAR



Terrestrial LiDAR



3D Pipeline



HD Map



CHAPTER 2

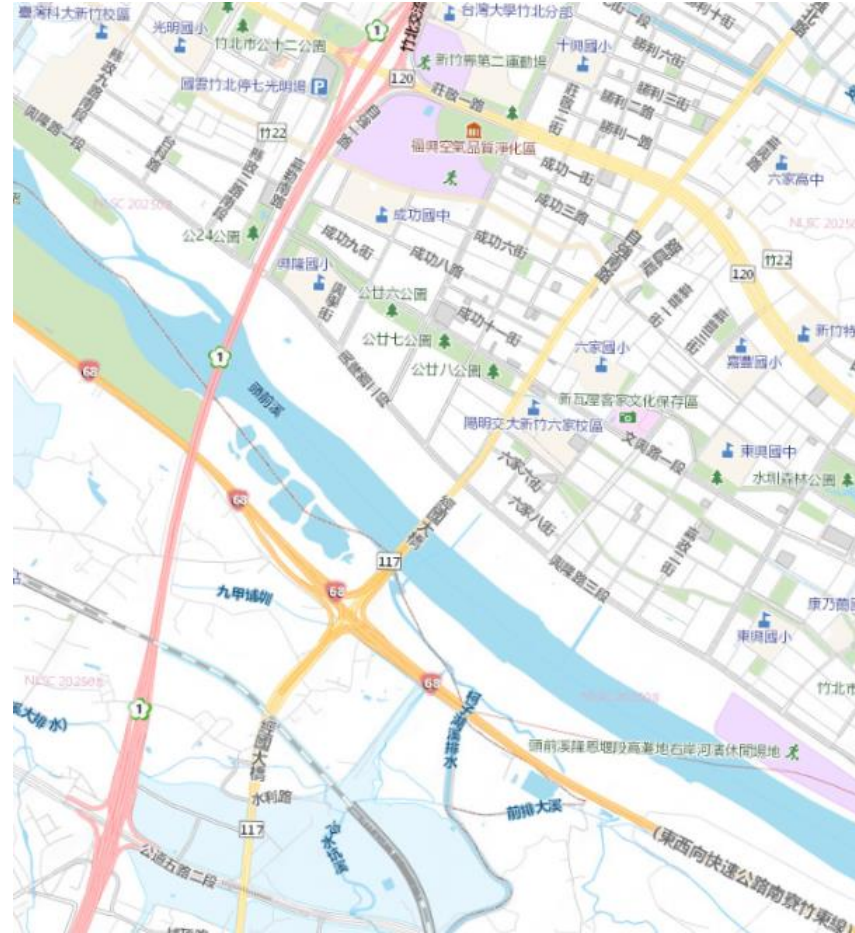
National Maps



- Since 2004
- Aerial Photogrammetry



Land Use Investigation



Taiwan Electric Map

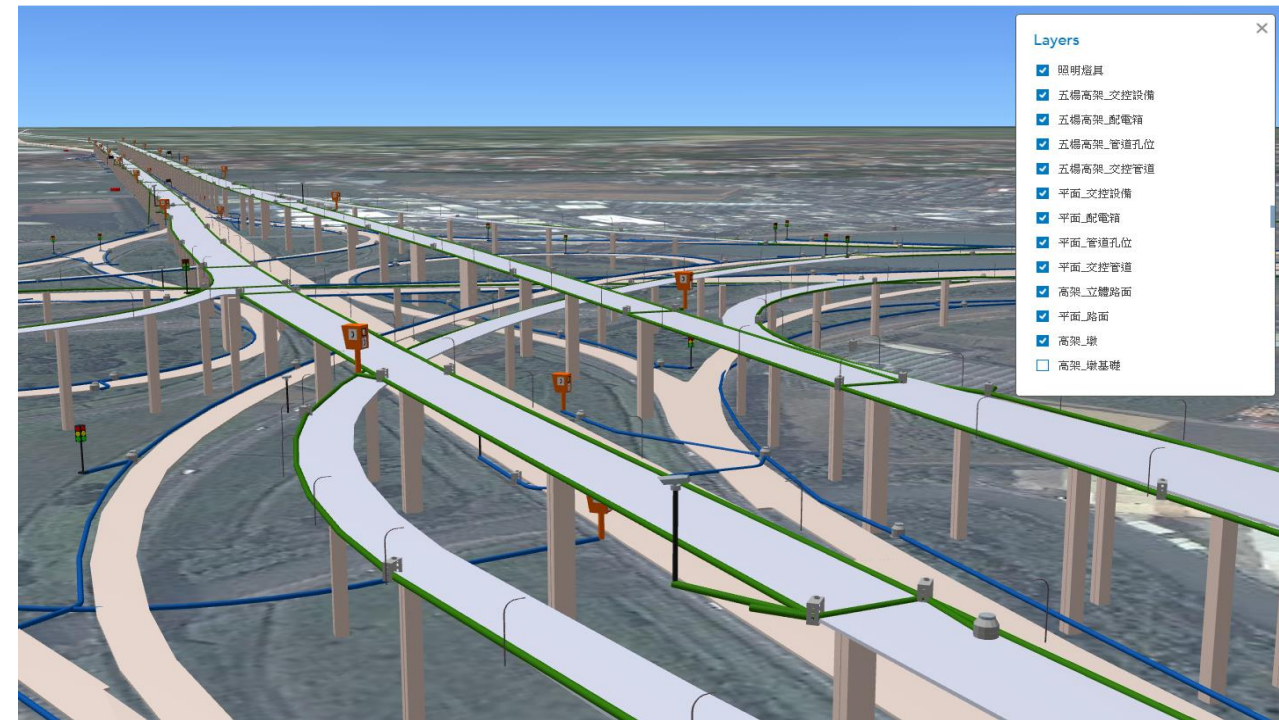


Basic Topographic Map

- Since 2019
- Emap + DEM + DSM + AI Texture Recognition
- Achieving 100% Coverage



3D Building Model (with Simulation Texture)



3D Road Model

-



- 2006-2023

Fomosa-2 Satellite
Launched



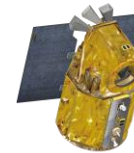
2004

2006

Decommissioning
of Fomosat-2
Satellite

2016

Fomosa-5 Satellite
Launched



2017

2022-
2024

Fomosa-8 Satellite
Launched

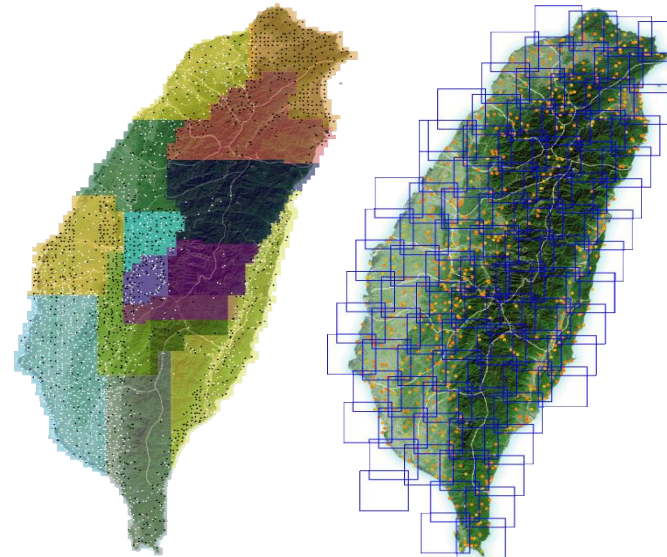


2025-
2028

GCF Project in 2006

Establishment, Drafting of Specifications and Supply Rules for Control Database of Aerial Photogrammetry and Remote Sensing

- Establishment of GCFs(4000pts)
- Formulation of Data Standards and Supply Methods
- Establishment of Circulation and Supply Systems

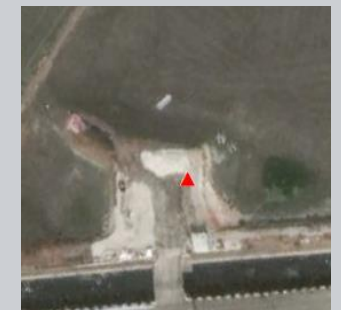


Distribution of GCFs derived from aerial and satellite images.

GCF Project in 2022-2024

Feature Point Database for Photogrammetry and Remote sensing images

The changes in the landscape are not consistent with the GCF project imagery from 2006.



CHAPTER 3

LiDAR Application





Total Station & Level



UAV & LiDAR

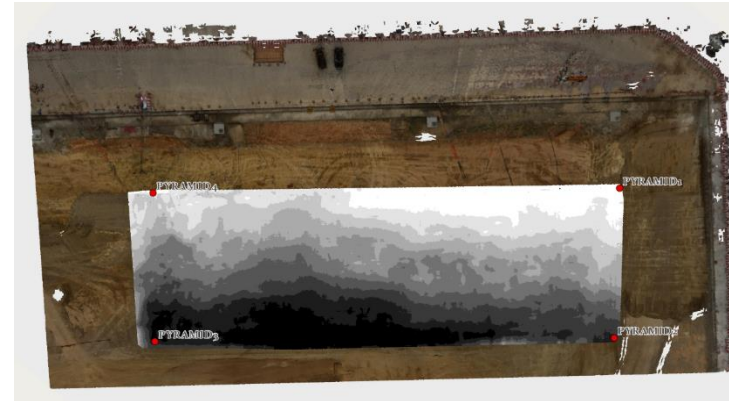


Photogrammetry
Point Cloud

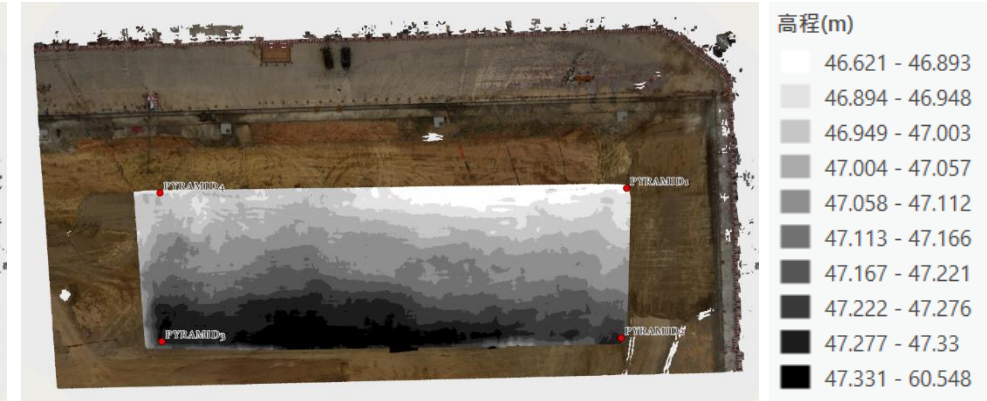
Wider Range, More Efficient, More Diversified Outcomes

◆ Inspection of the Finished Subgrade Surface

Allowable settlement < 25.4 mm



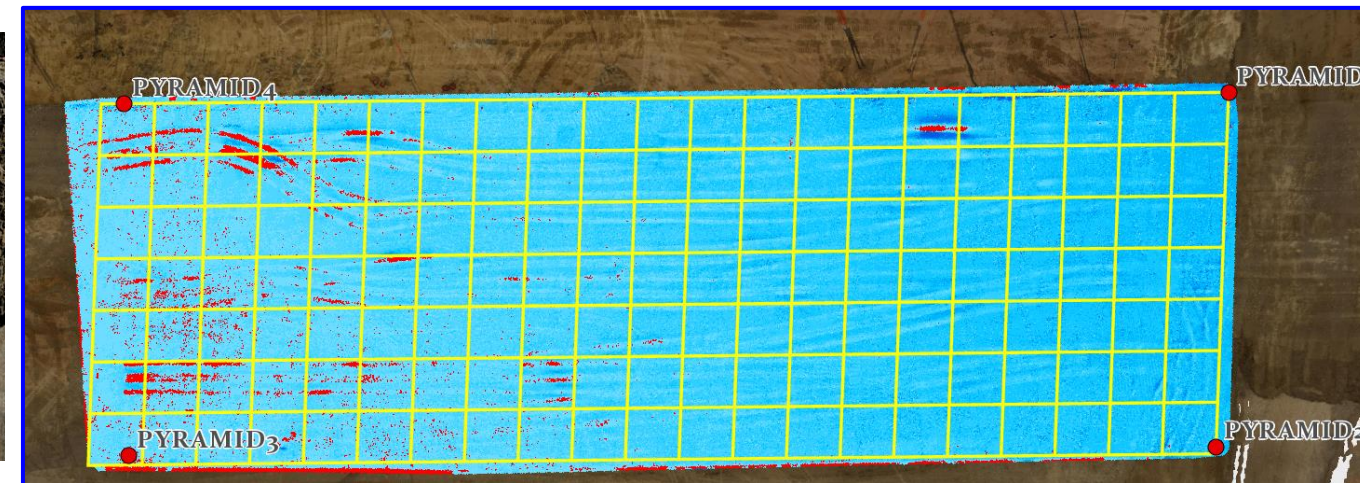
Before



After



Point Cloud Model



Red Area: settlement > 25.4mm

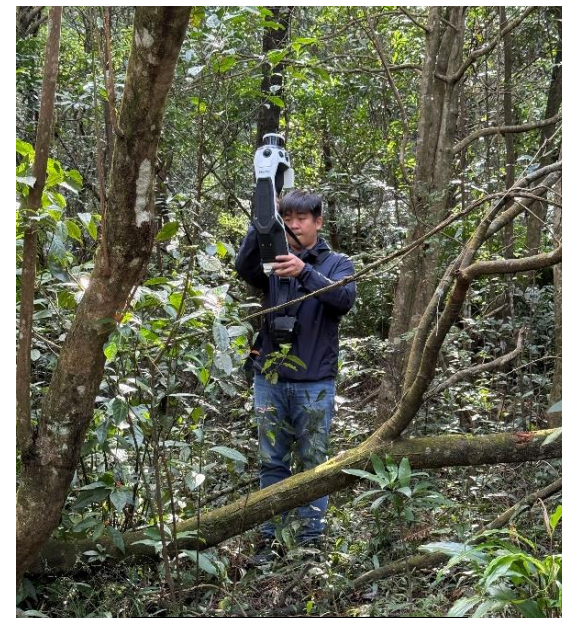
Comprehensive Area-Wide Settlement Monitoring

- **Forest trails**

- × **No GNSS Signal**
- × **Time-Consuming**
- × **Manpower-Intensive**

NavVis MLX Advantages

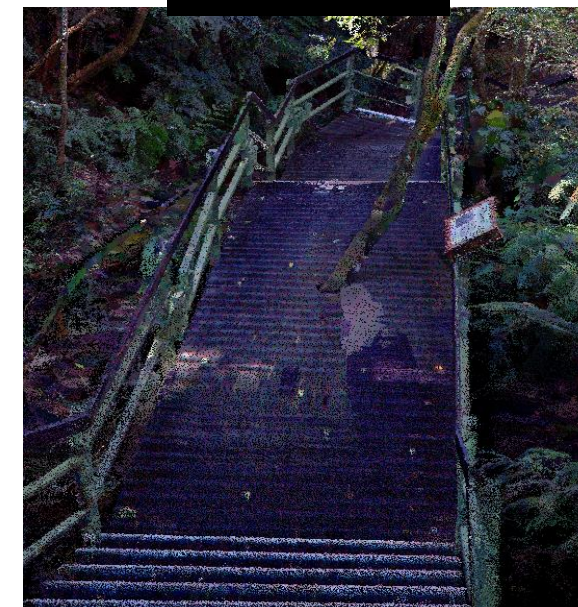
- ✓ **SLAM fast scanning**
- ✓ **Retain More Details**
- ✓ **3D Modeling**



Forest Scan



Trail Scan





NAVVIS MLX

大崙頭森林步道

CHAPTER 4

UAV Application



➤ Under 2kg

Visual Position
360° Sensing



Skydio X2

Stable
Long Battery Life



DJI Matrice 4E RTK

➤ Above 2kg



DJI M350 RTK



Airborne LiDAR
3D Terrain



Oblique Camera
3D Mesh Model



Phase One Camera
Stereo Mapping

➤ Above 15kg

NEW



Acecure NOA

- RTK
- Stable
- Metric camera
- RIEGL UAV LiDAR



UAV equipped with Phase One metric camera



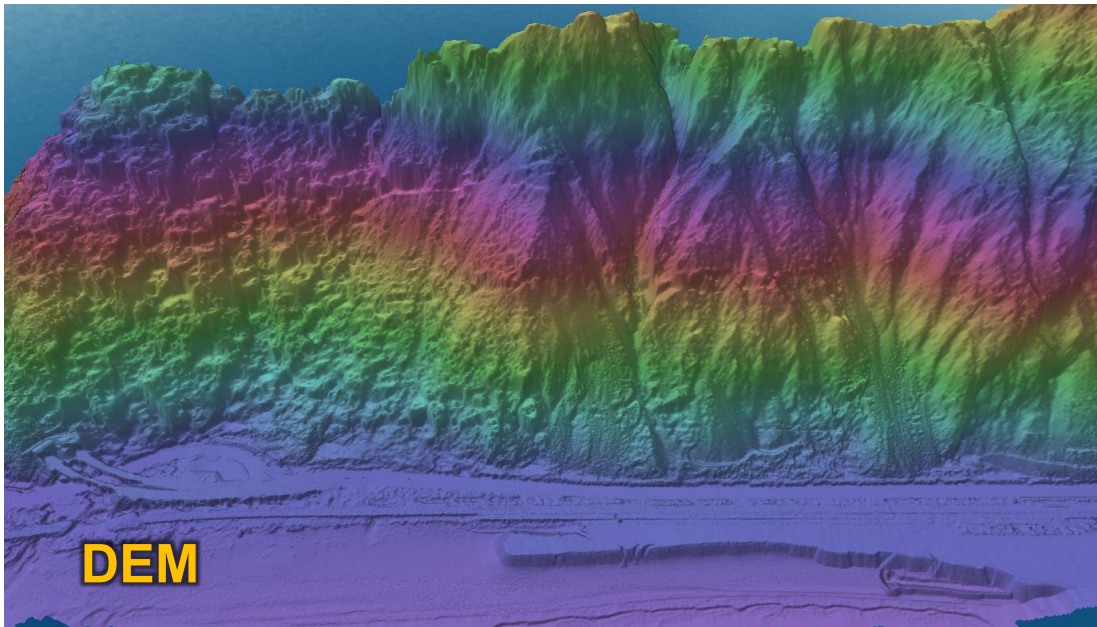
UAV Missions for Engineering Design in the Past Two Years

Project	Ha
Matai'an River Bridge	33
Expressway 62	170
Kaohsiung-Pingtung 2 nd Expressway	520
Suhua Safety	180
Tainan Science Park	340
Railway (North Link Line and Pingxi Line)	40

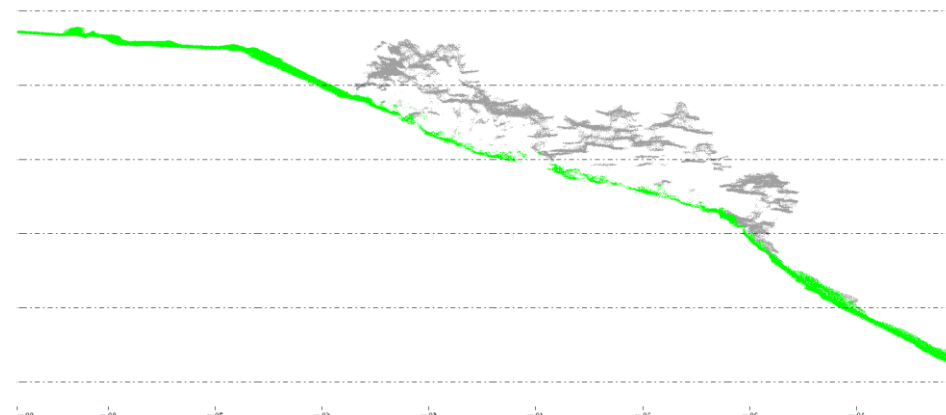
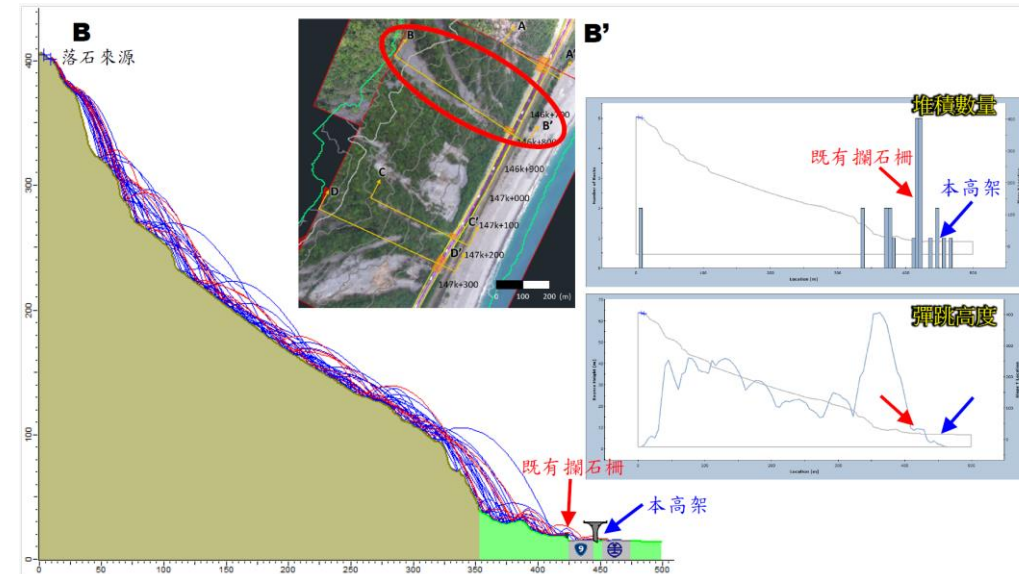
Camera	Phase One P3
Image Size	11,664 x 8,750 (100 megapixels)
Image Sensor Size	43.86 x 32.90mm
Pixel Size	3.76 μ m
Focal Length	35 mm
GSD@300m	3.22 cm
GSD@500m	5.37 cm
Coverage per Image	626 x 470 m
Weight	~2kg
Flight Time per Mission	~15min
GCP	Less



Suhua Safety



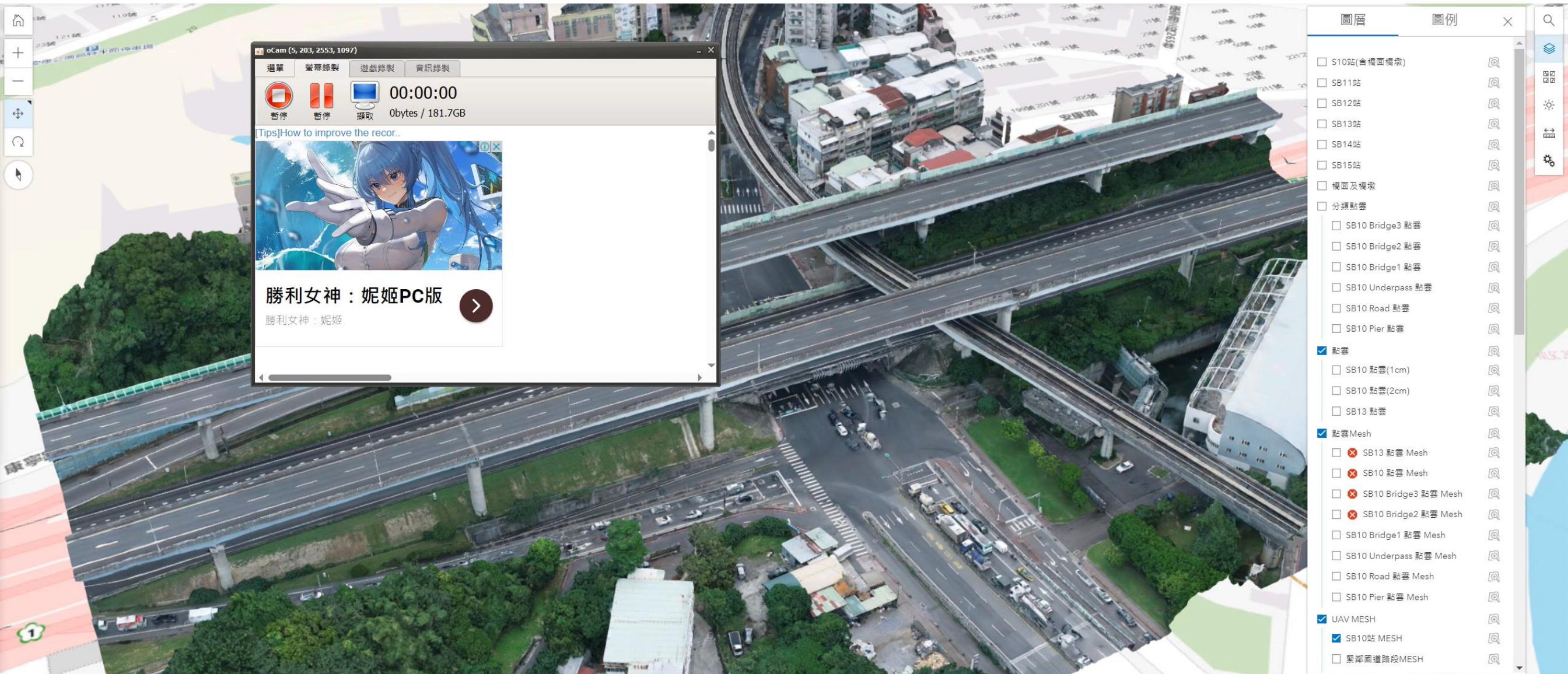
- Advantages of LiDAR over Image-Based Surveying
 - penetrate sparse vegetation
 - acquire data even at night

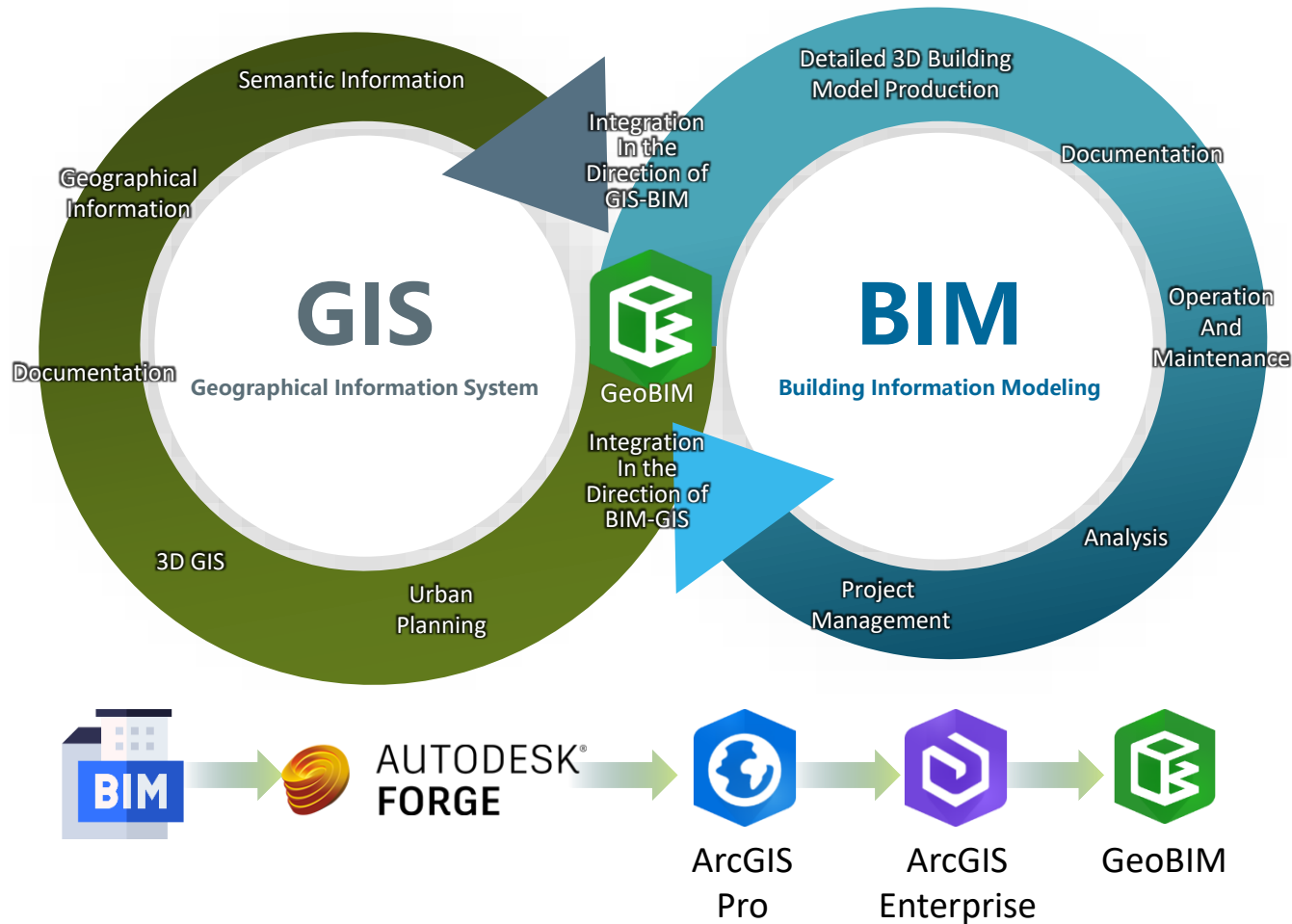


- Non-ground point
- ground point

LiDAR supports rockfall analysis in geologically fragile terrain

Scene Integration of Terrestrial LiDAR, UAV 3D Mesh, and Design Models





Digital Twins for Urban Planning and Design

花蓮地理資訊整合應用平台

首頁 網站導覽 資料瀏覽 服務申請 GIS圖台 子系統 New

test02 台灣世曦-空間資訊 登出 ? 操作手冊

3D 2D

圖台模式

三維地籍細緻模型

花蓮建物

- 秀林_台泥DAKA園區
- 新城_草林段都審周邊 Mesh
- 新城_煙波花蓮E館群樓 1
- 新城_煙波花蓮E館群樓 2
- 新城_煙波花蓮E館獨棟二次送件
- 新城_青年住宅BIM
- 花蓮_艾美酒店
- 花蓮_福祥段都審_邱召望建築師事務所
- 花蓮_富國段都審_劉燕湖建築師事務所
- 花蓮_富國段都審_局部
- 花蓮_雲門翠堤大樓
- 花蓮_花蓮機場禁限建範圍

瑞豐_都計區 Mesh 壽豐_都計區 Mesh 三維地籍_花蓮地所 三維地籍_美崙皇家 三維地籍_富野酒店 新城_草林段煙波花蓮都審 新城_煙波花蓮E館二次送件 花蓮_艾美酒店 花蓮市_福祥段都審

狀態: 屬性查詢 比例尺: 500
坐標 TWD97: X=314883, Y=2667928
目前介面: 電子地圖, 顯示比例尺: 1/1,000,000 ~ 1/200,000 by Esri

Source: USGS, NGA, NASA, CGIAR, GEBCO, N Robinson, NCEAS, NLS, OS, NMA, Geodastys, Esri and the GIS User Community

Powered by Esri

How can **“Geospatial Data”** be used in other disciplines?

- ◆ Integrating Surveying, GIS, and BIM to improve project management efficiency
- ◆ Transforming multi-source data into visualized and actionable outcomes
- ◆ Establishing a digital foundation for planning, construction, and operation

Thank You for Listening

Contact Us

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